Ashim Dey

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Faculty Profile | Google Scholar | LinkedIn | ResearchGate

Employment History

December, 2021 – Present — **Assistant Professor,** Department of Computer Science and Engineering, Chittagong University of Engineering and Technology (CUET), Bangladesh.

May, 2018 – December, 2021 — **Lecturer,** Department of Computer Science and Engineering, Chittagong University of Engineering and Technology (CUET), Bangladesh.

January, 2018 – May, 2018 — **Lecturer,** Department of Computer Science and Engineering, Green University of Bangladesh (GUB), Bangladesh.

Education

2025 — **M.Sc. Engg. in Computer Science and Engineering** *(Part-time)*
[Completed while working full-time as a faculty at the same university]
CGPA: **3.67 (with Honors)** / 4.00
Thesis title: Hybrid Caching with a Dynamic Forwarding Strategy in Information-Centric Network
Supervisor: Prof. Dr. Mohammad Shamsul Arefin
From: Chittagong University of Engineering and Technology (CUET), Bangladesh

2017 — **B.Sc. in Computer Science and Engineering**
CGPA: **3.78 (with Honors)** / 4.00 equivalent (**Merit position: 2nd** among 120)
Thesis title: Development of an ANN Model to Predict Road Traffic Congestion using Crowd Sourced GPS Data
Supervisor: Prof. Dr. Asaduzzaman
From: Chittagong University of Engineering and Technology (CUET), Bangladesh

Skills

Language Proficiency — Bengali - Native User
English - Proficient User (CEFR Level: C1)
IELTS Score - 7.0 (Listening-8.0|Reading-7.5|Writing-6.5|Speaking-6.5)

Coding — python, Java, C# (.net), SQL, XML/XSL, LaTeX, MATLAB

Misc. — Academic Research, Teaching, Training, Consultation, Mentoring, Reviewing

Research Publications

Journal Articles

1. S. A. Hanum, **A. Dey**, and M. A. Kabir, "An attention-guided deep learning approach for classifying 39 skin lesion types," *International Journal of Imaging Systems and Technology*, vol. 36, no. 1, e70269, 2026, e70269 IMA-25-1557.R1. DOI: https://doi.org/10.1002/ima.70269.

2. **A. Dey**, Asaduzzaman, M. S. Arefin, P. K. Dhar, and T. Shimamura, "A hybrid caching scheme with dynamic forwarding strategy in information-centric network," *IEEJ Transactions on Electrical and Electronic Engineering*, vol. 20, no. 6, pp. 920–933, 2025. DOI: https://doi.org/10.1002/tee.24251. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/tee.24251.

3. M. Osama, **A. Dey**, K. Ahmed, and M. A. Kabir, "Belin: A novel corpus for bengali religious news headline generation using contextual feature fusion," *Natural Language Processing Journal*, vol. 11, p. 100 138, 2025, ISSN: 2949-7191. DOI: https://doi.org/10.1016/j.nlp.2025.100138.

Conference Proceedings

1. A. Akhtar, **A. Dey**, A. M. Siddika, and A. Asaduzzaman, "Dp-spoof: Detection and prevention of arp spoofing using dynamic port and state allocation," in *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 2025, pp. 1–6. DOI: 10.1109/ECCE64574.2025.11013186.

2. A. Barua and **A. Dey**, "A temporal self-attention guided deep learning approach for prognosis of remaining useful life of aircraft turbofan engines," in *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 2025, pp. 1–6. DOI: 10.1109/ECCE64574.2025.11013861.

3. A. Barua, M. Osama, and **A. Dey**, "CUET_Novice@DravidianLangTech 2025: A Bi-GRU approach for multiclass political sentiment analysis of Tamil Twitter (X) comments," in *Proceedings of the Fifth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages*, B. R. Chakravarthi, R. Priyadharshini, A. K. Madasamy, *et al.*, Eds., Acoma, The Albuquerque Convention Center, Albuquerque, New Mexico: Association for Computational Linguistics, May 2025, pp. 496–501, ISBN: 979-8-89176-228-2. DOI: 10.18653/v1/2025.dravidianlangtech-1.87.

4. M. A. Chowdhury, L. T. Khan, M. S. Hasan, and **A. Dey**, "CUET_320@LT-EDI-2025: A multimodal approach for misogyny meme detection in Chinese social media," in *Proceedings of the 5th Conference on Language, Data and Knowledge: Fifth Workshop on Language Technology for Equality, Diversity, Inclusion*, K. Gkirtzou, S. Žitnik, J. Gracia, *et al.*, Eds., Naples, Italy: Unior Press, Sep. 2025, pp. 184–189, ISBN: 978-88-6719-334-9. URL: https://aclanthology.org/2025.ltedi-1.30/.

5. A. Z. U. Rahat, A. Das, and **A. Dey**, "Towards decentralizing the fashion industry: A blockchain-integrated system to mitigate dual pricing of branded apparel," in *2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, 2025, pp. 1–6. DOI: 10.1109/QPAIN66474.2025.11171920.

6. K. S. Sayma, F. A. Tofa, M. Osama, and **A. Dey**, "CUET_Novice@DravidianLangTech 2025: A multimodal transformer-based approach for detecting misogynistic memes in Malayalam language," in *Proceedings of the Fifth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages*, B. R. Chakravarthi, R. Priyadharshini, A. K. Madasamy, *et al.*, Eds., Acoma, The Albuquerque Convention Center, Albuquerque, New Mexico: Association for Computational Linguistics, May 2025, pp. 472–477, ISBN: 979-8-89176-228-2. DOI: 10.18653/v1/2025.dravidianlangtech-1.83.

7. F. A. Tofa, K. S. Sayma, M. Osama, and **A. Dey**, "CUET_Novice@DravidianLangTech 2025: Abusive comment detection in Malayalam text targeting women on social media using transformer-based models," in *Proceedings of the Fifth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages*, B. R. Chakravarthi, R. Priyadharshini, A. K. Madasamy, *et al.*, Eds., Acoma, The Albuquerque Convention Center, Albuquerque, New Mexico: Association for Computational Linguistics, May 2025, pp. 483–488, ISBN: 979-8-89176-228-2. DOI: 10.18653/v1/2025.dravidianlangtech-1.85.

8. F. A. Tofa, L. T. Zeba, M. Osama, and **A. Dey**, "CUET_INSights@NLU of Devanagari script languages 2025: Leveraging transformer-based models for target identification in hate speech," in *Proceedings of the First Workshop on Challenges in Processing South Asian Languages (CHiPSAL 2025)*, K. Sarveswaran, A. Vaidya, B. Krishna Bal, S. Shams, and S. Thapa, Eds., Abu Dhabi, UAE: International Committee on Computational Linguistics, Jan. 2025, pp. 267–272. URL: https://aclanthology.org/2025.chipsal-1.29/.

9. T. H. Tushi and **A. Dey**, "Towards smart diagnostic: A multi-disease prediction system leveraging machine learning techniques," in *2025 2nd International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*, 2025, pp. 1–6. DOI: 10.1109/NCIM65934.2025.11160008.

10. **A. Dey**, Asaduzzaman, and M. S. Arefin, "Cyhy: A centrality-aware hybrid caching strategy in information-centric networking," in *2024 IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON)*, 2024, pp. 1–6. DOI: 10.1109/SPICSCON64195.2024.10941074.

11. **A. Dey**, M. K. Islam, and S. Dhar, "Development of an iot-based pipe water quality monitoring and control system for smart city," in *2024 3rd International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE)*, 2024, pp. 1–6. DOI: 10.1109/ICAEEE62219.2024.10561879.

12. F. A. Khan and **A. Dey**, "Towards enhancing child safety: A deep learning approach to detect child safe and unsafe objects," in *2024 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)*, 2024, pp. 123–128. DOI: 10.1109/WIECON-ECE64149.2024.10914992.

13. S. I. Nova, A. Das, and **A. Dey**, "Multi-feature extraction-based phishing website detection using ensemble learning," in *2024 27th International Conference on Computer and Information Technology (ICCIT)*, 2024, pp. 861–866. DOI: 10.1109/ICCIT64611.2024.11021929.

14. M. M. R. Shuvo, **A. Dey**, A. S. Md.Toha, and M. O. Rahman, "An integrated approach of deeplabv3+ with an ensemble backbone for precise water bodies segmentation in remote sensing imagery," in *2024 27th International Conference on Computer and Information Technology (ICCIT)*, 2024, pp. 3641–3646. DOI: 10.1109/ICCIT64611.2024.11021916.

15. M. M. R. Shuvo, **A. Dey**, and M. O. Rahman, "A yolo-based framework for road sign detection and recognition in the context of bangladesh," in *2024 IEEE International Conference on Computing, Applications and Systems (COMPAS)*, 2024, pp. 1–6. DOI: 10.1109/COMPAS60761.2024.10796393.

16. F. S. Tamim, Z. S. Taheri, S. S. Niloy, **A. Dey**, and A. Das, "An ensemble approach of transfer learning and vision transformer to identify covid-19 from chest x-rays," in *2024 3rd International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE)*, 2024, pp. 1–6. DOI: 10.1109/ICAEEE62219.2024.10561636.

17. M. Al Shahriar and **A. Dey**, "A hybrid approach of cnn and lstm to detect intrusion in edge iot devices using catboost," in *2023 26th International Conference on Computer and Information Technology (ICCIT)*, 2023, pp. 1–6. DOI: 10.1109/ICCIT60459.2023.10441595.

18. T. Ibnath and **A. Dey**, "Toward a transfer learning approach to detect face mask type in real-time," in *2023 3rd International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST)*, 2023, pp. 352–357. DOI: 10.1109/ICREST57604.2023.10070040.

19. A. Banik, A. Das, and **A. Dey**, "Toward developing a password breach detection technique for serving the purpose of honeywords," in *2022 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)*, 2022, pp. 66–71. DOI: 10.1109/WIECON-ECE57977.2022.10151354.

20. S. D. Bappon, **A. Dey**, S. M. Sabuj, and A. Das, "Toward a machine learning approach to predict the co2 rating of fuel-consuming vehicles in canada," in *2022 25th International Conference on Computer and Information Technology (ICCIT)*, 2022, pp. 384–389. DOI: 10.1109/ICCIT57492.2022.10054732.

21. S. Ara, A. Das, and **A. Dey**, "Malignant and benign breast cancer classification using machine learning algorithms," in *2021 International Conference on Artificial Intelligence (ICAI)*, 2021, pp. 97–101. DOI: 10.1109/ICAI52203.2021.9445249.

22. R. Faria, F. Ahmed, A. Das, and **A. Dey**, "Classification of organic and solid waste using deep convolutional neural networks," in *2021 IEEE 9th Region 10 Humanitarian Technology Conference (R10-HTC)*, 2021, pp. 01–06. DOI: 10.1109/R10-HTC53172.2021.9641560.

23. A. I. Hossain, S. Sikder, A. Das, and **A. Dey**, "Applying machine learning classifiers on ecg dataset for predicting heart disease," in *2021 International Conference on Automation, Control and Mechatronics for Industry 4.0 (ACMI)*, 2021, pp. 1–6. DOI: 10.1109/ACMI53878.2021.9528169.

24. A. S. Nasim, R. H. Chowdory, **A. Dey**, and A. Das, "Recognizing speech emotion based on acoustic features using machine learning," in *2021 International Conference on Advanced Computer Science and Information Systems (ICACSIS)*, 2021, pp. 1–7. DOI: 10.1109/ICACSIS53237.2021.9631319.

25. M. G. Sarwar, **A. Dey**, and A. Das, "Developing a lbph-based face recognition system for visually impaired people," in *2021 1st International Conference on Artificial Intelligence and Data Analytics (CAIDA)*, 2021, pp. 286–289. DOI: 10.1109/CAIDA51941.2021.9425275.

26. R. Tasnim, S. T. Pritha, A. Das, and **A. Dey**, "Bangladeshi banknote recognition in real-time using convolutional neural network for visually impaired people," in *2021 2nd International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST)*, 2021, pp. 388–393. DOI: 10.1109/ICREST51555.2021.9331182.

27. S. Chowdhury, S. Nath, **A. Dey**, and A. Das, "Development of an automatic class attendance system using cnn-based face recognition," in *2020 Emerging Technology in Computing, Communication and Electronics (ETCCE)*, 2020, pp. 1–5. DOI: 10.1109/ETCCE51779.2020.9350904.

28 M. N. Islam Opu, T. K. Koly, A. Das, and **A. Dey**, "A lightweight deep convolutional neural network model for real-time age and gender prediction," in *2020 Third International Conference on Advances in Electronics, Computers and Communications (ICAECC)*, 2020, pp. 1–6. DOI: 10.1109/ICAECC50550.2020.9339503.

29 A. Pondit, **A. Dey**, and A. Das, "Real-time driver monitoring system based on visual cues," in *2020 6th International Conference on Interactive Digital Media (ICIDM)*, 2020, pp. 1–6. DOI: 10.1109/ICIDM51048.2020.9339604.

Books and Chapters

1 S. Akter, N. Nabila, **A. Dey**, and A. Das, "Analyzing the it job market and classifying it jobs using machine learning algorithms," in *Applied Intelligence for Industry 4.0*, Chapman and Hall/CRC, May 2023, pp. 240–252, ISBN: 9781003256083. DOI: 10.1201/9781003256083.

2 A. Das, Asaduzzaman, and **A. Dey**, "Analyzing the errors in channel sensing and negotiation in cognitive radio h-cran," in *Edge Analytics*, Springer Singapore, 2022, pp. 437–451, ISBN: 9789811900198. DOI: 10.1007/978-981-19-0019-8_33.

3 M. M. Hasan, S. Islam, **A. Dey**, A. Das, and S. C. Tista, "Detection of malaria disease using image processing and machine learning," in *Intelligent Computing & Optimization*, Springer International Publishing, 2022, pp. 457–466, ISBN: 9783030932473. DOI: 10.1007/978-3-030-93247-3_45.

4 S. I. Rimon, M. R. Islam, **A. Dey**, and A. Das, "Plantbuddy: An android-based mobile application for plant disease detection using deep convolutional neural network," in *Artificial Intelligence and Technologies*, Springer Singapore, Dec. 2021, pp. 275–285, ISBN: 9789811664489. DOI: 10.1007/978-981-16-6448-9_28.

5 M. Zubair, S. Ahmed, **A. Dey**, A. Das, and M. M. Hasan, "An intelligent model to suggest top productive seasonal crops based on user location in the context of bangladesh," in *Smart Systems: Innovations in Computing*, Springer Singapore, Sep. 2021, pp. 289–300, ISBN: 9789811628771. DOI: 10.1007/978-981-16-2877-1_26.

Miscellaneous Experience

Certification

QSilver DIPLOMA — Intermediate level workshop series by QWorld on quantum computing and programming focusing on complex numbers, QFT and Shor's Algorithm quantum computing and programming using Qiskit **(2025)**

QBronze DIPLOMA — Introductory level quantum workshop series (16 to 20 hours) by QWorld on the basics of quantum computing and quantum programming using Qiskit **(2025)**

Qiskit Global Summer School 2025 — Qiskit Global Summer School (QGSS) is a 2-week, fully virtual event designed to guide students through the evolution of quantum technologies using Qiskit by IBM **(2025)**

Qiskit Fall Fest Bangladesh 2025 — The first-ever Qiskit Fall Fest Extension event in Bangladesh by IBM Quantum. The focus is on fostering community, learning, and collaboration around Qiskit, IBM's open-source quantum software, and quantum computing as a whole. **(2025)**

Hands on Training on Research — by Institutional Quality Assurance Cell (IQAC),Chittagong University of Engineering & Technology (CUET) **(2024)**

Training on Outcome Based Education — by Institutional Quality Assurance Cell (IQAC),Chittagong University of Engineering & Technology (CUET) **(2024)**

60 hrs training on Fundamental of IoT — by ICT Division, Govt. of Bangladesh **(2020)**

Foundation Training for Teachers — by Institutional Quality Assurance Cell (IQAC),Chittagong University of Engineering & Technology (CUET) **(2020)**

Miscellaneous Experience (continued)

Organizational Activities

Moderator (2024-2025)
- **CUET Model United Nations Club**, an educational simulation of the United Nations to teach students about diplomacy, international relations, and global issues.

Moderator (2018-Present)
- **CUET Computer Club**, a club to promote the consciousness of information technology among students
- **Green For Peace, CUET**, an organization of CUET for consciousness about nature conservation and pollution
- **Greater Mymensingh Association, CUET**, an regional organization

Member (Finance Sub-committee)
- **IEEE 4th International Conference on Electrical, Computer and Communication Engineering (ECCE 2025)**

Member (Organizing committee)
- **1st National Workshop on Big Data and Machine Learning (BDML 2020), CUET**

Course Co-ordinator (2023-2025)
- **2019 Batch, CSE, CUET**, Responsibility includes Mentoring particular batch students and offering them guidance in course selection, career paths, and academic goals, as well as research

Experience

Reviewer (Journal)
- Journal of Data in Brief, Elsevier
- Journal of Heliyon, Elsevier
- Journal of Informatics in Medicine Unlocked, Elsevier

Reviewer (Conference)
- IEEE 3rd International Conference on Big Data, IoT and Machine Learning (BIM 2025), Bangladesh
- IEEE 2nd International Conference on Computing Applications and Systems (COMPAS 2025), Bangladesh
- IEEE 4th International Conference on Electrical, Computer and Communication Engineering (ECCE 2025), Bangladesh
- IEEE 3rd International Conference on Intelligent Systems, Advanced Computing, and Communication (ISACC 2025), India
- Fifth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages (DravidianLangTech-2025) at NAACL 2025, New Mexico.
- Springer International Conference on Data Science, Artificial Intelligence and Applications (ICDSAIA 2025), Bangladesh.
- Fifth Workshop on Language Technology for Equality, Diversity, Inclusion (LT-EDI-2025) at LDK 2025, Italy.

Coach
- Coached and mentored teams for various prestigious programming competitions, including ICPC Asia Dhaka Regional 2023-2025, BUET IUPC 2026-23 DUET IUPC 2025, AUST IUPC 2025, IUT IUPC 2024, SUST IUPC 2024, KUET IUPC 2025, CoU-BRACNet IUPC 2023, National Girls Programming Contest (NGPC) 2022 and others.

Judge
- Contributed as a Judge for the Final Round of the poster presentation competition at SciBlitz 1.0, organized by the IEEE CUET Student Branch

Miscellaneous Experience (continued)

Session Moderator
- IEEE 4th International Conference on Electrical, Computer and Communication Engineering (ECCE 2025), Bangladesh

Awards and Achievements

Academic Merit Award
- Received the Academic Merit Award for outstanding academic performance consecutively from 2013 to 2017, spanning 8 semesters during B.Sc.

ACM-ICPC 2014 Asia Regional
- Participated in the Dhaka Site Online Preliminary Contest which was sponsored by IBM and Hosted by BUBT

International Programming Contest 2014
- Participated in Quazi Azher Ali International Programming Contest 2014, hosted by Bangladesh University

Extracurricular Activities

Organizing Secretary (2016-2017)
- **Green For Peace, CUET**, an organization of CUET for consciousness about nature conservation and pollution

Finance and Office Secretary (2016-2017)
- **Moshal**, a charity organization of CUET

Training and Workshops

LICT TOP-UP Dot NET
- 300hrs (60 hrs soft skills + 240 hrs technical skills) training under NASSCOM IT-ITES Sector Skill Council (SSC) Certified by George Washington University, USA (2017)

Android Application Development
- 3 Months Short course by IICT, CUET (2017)

Mobile Game Graphics Design
- 40 Hours training by ICT Division, Bangladesh(2017)

Advanced Mobile Game Development
- 40 Hours training by ICT, Division Bangladesh(2017)

Cloud Computing & NoSQL Databases
- Daylong workshop organized by CSE, CUET (2017)

Career & Employment
- Daylong workshop organized by CUET Career Club (2017)

Industrial Trainee
- 15 days industrial attachment at MINZ IT, Bangladesh (2017)

References

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